Retail Pricing & Demand Forecasting

Dataset Analysis Questions Guide

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Datasets: Rossmann Store Sales | Retail Inventory | ISTAT Inflation

This document defines the specific business questions you should answer from each dataset in this project. For every question, the relevant columns and the business reason are clearly stated. Focus your cleaning and analysis only on the columns listed here — ignore everything else.

Dataset 1 — Rossmann Store Sales

File: 01_kaggle_rossmann/rossmann_store_sales_train.csv | 1,017,209 rows | 1,115 stores | Jan 2013 – Jul 2015

Q1. Which store type generates the highest average daily sales?

Columns: StoreType → Sales

Why it matters: *Rossmann has 4 store types (a, b, c, d). Knowing which layout sells most helps justify store format decisions on your dashboard.*

Q2. Does running a promotion increase sales, and by how much?

Columns: Promo → Sales, Customers

Why it matters: *Core KPI for any retail dashboard. Quantify the revenue uplift from promotions — this directly feeds your 'Promotion Effectiveness' dashboard panel.*

Q3. What is the monthly sales trend? Is there a seasonal pattern?

Columns: Date (extract Month/Year) → Sales

Why it matters: *Seasonality drives forecasting. Spotting peaks (e.g. December) and troughs validates your demand model and is a key chart for any portfolio project.*

Q4. Do public holidays hurt or boost sales?

Columns: StateHoliday → Sales

Why it matters: *StateHoliday has values: 0 (none), a (public), b (Easter), c (Christmas). Comparing sales across these reveals how the Italian retail calendar affects revenue.*

Q5. Does competition proximity affect a store's revenue?

Columns: CompetitionDistance → Sales

Why it matters: *Bin CompetitionDistance into ranges (0-500m, 500m-2km, 2km+) and compare average Sales. A key strategic insight for store placement decisions.*

Q6. Which product assortment type drives highest customer footfall?

Columns: Assortment → Customers

Why it matters: *Assortment types: a (basic), b (extra), c (extended). More customers does not always mean more sales — this comparison surfaces the quality vs. volume trade-off.*

Q7. What percentage of store-days have zero sales, and why?

Columns: Open → Sales

Why it matters: *Open=0 means the store was closed. Filtering these out before computing averages is critical — including closed days artificially deflates all your sales metrics.*

Dataset 2 — Retail Inventory Sales

File: 02_kaggle_demand_forecasting/retail_inventory_sales.csv | 76,000 rows | Jan 2022 – Jan 2024

Q1. Does a higher discount percentage lead to more units sold?

Columns: Discount → Units Sold

Why it matters: *This is price elasticity analysis. Group Discount into bands (0%, 1-10%, 11-20%, 20%+) and compare mean Units Sold. The core question of any pricing project.*

Q2. Which product category has the highest average demand?

Columns: Category → Demand, Units Sold

Why it matters: *Identifies your top-performing product lines. Essential for the 'Revenue & Units Sold' KPI panel and for prioritising inventory planning.*

Q3. How does our price compare to competitor pricing, and does the gap affect sales?

Columns: Price, Competitor Pricing → Units Sold

Why it matters: *Create a column: price_gap = Price - Competitor Pricing. Negative gap (we are cheaper) should correlate with higher Units Sold. This is the strategic pricing insight.*

Q4. Does running a promotion significantly boost demand?

Columns: Promotion → Demand, Units Sold

Why it matters: *Directly maps to your 'Promotion Effectiveness' dashboard KPI. Compare mean Demand when Promotion=1 vs Promotion=0 to get a clear uplift percentage.*

Q5. Which season drives the most sales volume?

Columns: Seasonality → Units Sold, Demand

Why it matters: *Seasonality column has values: Winter, Spring, Summer, Autumn. A grouped bar chart of this is one of the most visually clear insights for a portfolio dashboard.*

Q6. What is the relationship between price and demand across categories?

Columns: Price, Category → Demand

Why it matters: *Some categories are price-sensitive (elastic) and others are not (inelastic). A scatter plot of Price vs Demand faceted by Category is a standout EDA visualisation.*

Dataset 3 — ISTAT Inflation (HICP / NIC)

Files: 03_istat_inflation/istat_hicp_monthly_4digit.csv & istat_nic_weights_6digit.csv | Italy CPI 2015 – 2026

Q1. What is the overall inflation trend in Italy from 2015 to 2025?

Columns: TIME_PERIOD → Observation (filter: ECOICOP_2 = 'CP00' overall index)

Why it matters: *This is the headline number. Plot monthly HICP index over time to show when Italy experienced high inflation (2022-2023 energy crisis) vs stability.*

Q2. Which retail product categories experienced the steepest price increases?

Columns: ECOICOP 2 (category name) → Observation (index value), TIME_PERIOD

Why it matters: *Compare CPI index values across food, clothing, household goods categories. This tells you which categories are most inflation-sensitive — key context for your pricing analysis.*

Q3. How do you compute inflation-adjusted price from raw sales data?

Columns: TIME_PERIOD + Observation → joined to sales data on year-month

Why it matters: *Formula: adjusted_price = nominal_price / (CPI_t / CPI_base_year). This is the engineered feature that connects the ISTAT data to your Rossmann and inventory datasets.*

Q4. Does the HICP weight of a product category predict how elastic its demand is?

Columns: ECOICOP 2, Observation (NIC weights) → cross-reference with retail Discount vs Demand

Why it matters: *Higher-weighted necessities (food, housing) tend to be inelastic. Lower-weighted discretionary items tend to be elastic. This is an advanced insight that elevates the project.*

Dataset 4 — Eurostat CPI

Files: 04_eurostat_cpi/eurostat_hicp_annual_index.csv & eurostat_hicp_inflation_rate.csv | Italy 2010 – 2025

Q1. What was Italy's annual inflation rate year by year from 2014 to 2025?

Columns: TIME_PERIOD → OBS_VALUE (filter geo = Italy)

Why it matters: *Use tec00118 inflation rate file. This gives you the % change per year — a clean time series for your dashboard context panel showing macro environment.*

Q2. How does Italy's inflation compare to the EU average?

Columns: geo, TIME_PERIOD → OBS_VALUE (compare Italy vs EU27)

Why it matters: *The HICP annual index file contains all EU countries. A comparison line chart (Italy vs EU average) shows whether Italian retail operates in a high or low inflation environment relative to peers.*

Columns you can safely ignore across all datasets

Dataset	Columns to Ignore	Reason
Rossmann	CompetitionOpenSinceMonth, CompetitionOpenSinceYear, Promo2SinceWeek, Promo2SinceYear, PromoInterval, DayOfWeek, SchoolHoliday	Promo2 detail is redundant with Promo flag. DayOfWeek duplicates Date extraction. SchoolHoliday low signal for Italy analysis.
Retail Inventory	Store ID, Product ID, Inventory Level, Units Ordered, Weather Condition, Epidemic, Region	Store/Product IDs are codes with no analytical value. Weather and Epidemic have too few distinct values to drive meaningful insight.
ISTAT / Eurostat	FREQ, Frequency, REF_AREA, DATA_TYPE, MEASURE, OBS_STATUS, Observation status	These are metadata columns describing the data format, not the data itself. Only TIME_PERIOD, ECOICOP 2, and Observation matter.